Sachin Saailesh Jeyakkumaran

+1 (716) 939-8026 | sachin.jeyy@gmail.com | linkedin.com/in/sachinjeyakkumaran | GitHub | Portfolio

Summary

AI Engineer with 3+ years building distributed AI systems, real-time inference pipelines, and production RAG platforms across edge and cloud. Experienced in model deployment, evaluation-driven optimization, and MLOps using FastAPI, Docker, and PyTorch. Delivered AI systems across video analytics, healthcare AI, voice AI and legal intelligence.

Technical Skills

Languages: Python, C++, SQL, Java, JavaScript, Node.js, Bash, Git

AI/ML Frameworks: PyTorch, TensorFlow, ReAct, Transformers, BERT, LangGraph, CrewAI, Google ADK, OpenAI, Deepgram, ElevenLabs

Model Development: LLM Fine-tuning (LoRA, QLoRA, RLHF, DPO), Prompt Engineering, Function/Tool Calling, Pydantic, Instructor

Evaluation & Optimization: LLM-as-Judge Evaluation, LangSmith, Ragas/DeepEval Metrics, A/B Testing, Lumoz AI, Groundcover

MLOps & Deployment: FastAPI, Docker, Kubernetes, Airflow, MLflow, CI/CD, NVIDIA Triton, vLLM, TensorRT, Prometheus, OpenTelemetry

LLM & Retrieval Systems: RAG, Vector Databases (Pinecone, Qdrant, ChromaDB), Dialogflow, Multi-step Agents, Cohere Rerank, Neo4j

Cloud & Data: AWS (S3, Lambda, ECR, Bedrock), Azure, GCP, Vertex AI, LiveKit, Redis, Kafka, PostgreSQL, Datadog, Terraform, Twilio

Education

State University of New York at Buffalo **Aug 2024 – May 2026**
Master of Science, Artificial Intelligence

Sri Ramachandra Faculty of Engineering and Technology **Aug 2019 – May 2023**
Bachelor of Technology, Computer Science & Engineering (AI & ML)
Teaching Assistant: Data Visualization, Computer Networks, Ethical Hacking

Experience

Tabhi | *AI Builder* **May 2026 – Present**

- Launched TACO (Tabhi Agentic Call-CenterOps), a voice agent fielding travel-disruption calls over Twilio PSTN trunking into LiveKit with Soniox STT, Gemini Flash inference, Cartesia TTS, and Google ADK orchestration at 500ms voice-to-voice latency.
- Automated airline rebooking, ticket reissue, and hotel and restaurant reservations by resolving caller PNRs against airline GDS and partner APIs; idempotency keys and spoken confirmation gated every fare-difference charge.
- Provisioned GKE clusters, LiveKit media servers, and Twilio SIP trunks via Terraform; autoscaled ADK agent pods on concurrent-session load with graceful draining, sustaining 100 concurrent calls at peak without dropping sessions.
- Instrumented groundcover monitoring across STT→LLM→TTS spans with alerts on turn latency, call drops, and tool-error rate; cascading TTS vendor failover and warm transfer to live agents cut abandoned calls 18% during outages.
- Raised call containment from 52% to 68% by mining failed-call transcripts weekly with ops to reprioritize intents and prompts; cut cost per resolved call 21% via model routing and TTS caching of static turns.

Rebecca Everlene Trust Company | *AI Engineer* **Feb 2026 – May 2026**

- Owned end-to-end delivery of LLM features for enterprise HCM clients, from requirements discovery to production, on FastAPI, Google ADK, and AWS Bedrock; shipped 6+ workflows across 4 deployments, cutting manual processing 35%.
- Engineered entitlement-aware RAG over HR, payroll, and compliance chunks via hybrid BM25/dense Pinecone retrieval, cross-encoder reranking, and jurisdiction and permission filters; lifted Recall@5 24%, cut ungrounded answers 30%.
- Built agentic workflows in Google ADK with tool calling and MCP into AWS Lambda, S3, and client APIs, gating payroll writes behind human approval callbacks; idempotent retries and durable session state eliminated double-posting.
- Established eval and traceability layer with Lumoz eval suites on every prompt/model change, LLM-as-judge groundedness scoring, and OpenTelemetry tracing; delivered per-decision audit trails under NYC LL 144 and Illinois HB 3773.

PiSence Technologies | *AI Engineer* **Aug 2023 – Jul 2024**

- Architected end-to-end distributed video analytics platform on AWS SageMaker, handling 100+ concurrent camera streams at 30 FPS with real-time edge deployment for enterprise-scale activity recognition and compliance.
- Engineered async FastAPI microservice backed by Triton Inference Server sustaining sub-150ms latency across 100+ concurrent streams with resilient fallback routing and graceful degradation for uninterrupted production serving.
- Productionized predictive maintenance pipelines using LSTM/XGBoost with MLflow and Evidently AI drift detection, achieving 24-48 hour failure prediction and 28% operational downtime reduction across 10M+ daily sensor readings.
- Deployed anomaly and intrusion detection over 50M+ daily logs using Isolation Forest and Autoencoders; instrumented Prometheus/OpenTelemetry dashboards monitoring model drift, latency, and pipeline health in real-time.

Calidad Healthcare | *AI/ML Engineer* **Jun 2022 – Jul 2023**

- Built HIPAA-compliant multi-modal AI platform across 4 imaging modalities with Dockerized FastAPI, serving 2K+ daily clinical predictions at 99.5% uptime with full audit trails meeting clinical governance and regulatory compliance standards.
- Partnered with radiologists and clinical staff across 3 hospital sites to define accuracy thresholds and translate clinical workflows into model eval criteria; cut diagnostic errors 35% through explainability and iterative refinement.
- Trained CNN and transformer models in PyTorch/TensorFlow with focal loss, class balancing, and adaptive augmentation; governed experiments via MLflow enforcing reproducibility and accelerating model versioning and deployment cycles.
- Delivered clinical admin demos translating Grad-CAM and SHAP outputs into plain narratives; accelerated deployment approvals by 3 weeks per site through clear communication of model explainability and clinical safety.

Projects

Maya: Real-Time Voice AI Event Planner (Multi-Step Agent) | *Python, FastAPI, ChromaDB, LiveKit, Whisper, React* ***GitHub***

- Orchestrated end-to-end voice agent on LiveKit+Whisper resolving 12 conversation slots with <2% slot-fill error and 91% intent accuracy; built QA lab with 38 test scenarios tracking STT→LLM→TTS latency and detecting regressions automatically.
- Reduced hallucination rate 43% via ChromaDB RAG, semantic caching, and LLM-as-judge eval on 200-sample set; iterated 8 prompt versions via LangSmith, cut API calls 38%, maintained <1.6s E2E latency with 100% production QA pass rate.

LexMind: Multi-Agent Legal Intelligence Platform | *Python, LangGraph, FastAPI, Qdrant, Neo4j, Azure OpenAI, Next.js* ***GitHub***

- Architected LangGraph orchestrator coordinating 5 specialist agents (contract, compliance, litigation, drafting, research) over 4-stage hybrid RAG (Qdrant+BM25 via RRF) cutting manual legal review time 60%.
- Engineered Hallucination Guard routing GPT-4o outputs <70% confidence to human review; collected attorney corrections as high-weight chunks improving retrieval; designed Ragas evaluation framework tracking context precision on 80+ test cases.